Math interlude

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1 Matrix exponential

For the simple exponential

$$\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \\
&= \lim_{n \to \infty} (1 + x/n)^n \\
e^x e^y &= e^{x+y} \\
\frac{\partial e^x}{\partial x} &= e^x
\end{aligned}$$

For a matrix A

$$\begin{aligned}
e^A &= I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \ldots \\
&= \lim_{n \to \infty} (I + A/n)^n \\
e^A e^B &\neq e^{A+B} \\
\frac{\partial e^A}{\partial A_{ij}} &= ?
\end{aligned}$$

The matrix exponential is a lot more difficult to compute; matrices where the elements of A are quite different is size are challenging. For a 9 state model the code below shows that the matrix exponential computation takes on the order of 30 times an ordinary exponential.

```
A <- matrix(1:81, 9, byrow=TRUE)/200
diag(A) <- diag(A)- rowSums(A)  # make the rows sum to 0
system.time({for (i in (1:1e5)) exp(A)})

##    user  system elapsed
##   0.048   0.008   0.055

system.time({for (i in (1:1e5)) expm(A)})

##    user  system elapsed
##   1.447   0.000   1.448
```

Relation to survival

$N_i(t)$	cumulative number of events for subject i, up to time t
$N_{ijk}(t)$	in a multistate model, the number of transitions from state j to state k for subject i
$Y_i(t)$	0/1 indicator that subject i is at risk at time t
$Y_{ij}(t)$	in a multistate model, the indicator that subject i is at risk for a transition out of state j
X	n by p matrix of predictors where p is the number of predictors, X_i the row vector for observation i
w	vector of case weights for the subjects, W a diagonal matrix of the weights
β	vector of estimates
$\eta = X\beta$	vector of linear predictors

The most common estimates of the cumulative hazard and survival are the Nelson-Aalen and Kaplan-Meier estimates, sometimes Fleming-Harrington

$$\begin{aligned}
\hat\lambda(t) &= \frac{\sum w_i dN_i(s)}{\sum_i w_i Y_i(s)} \\
\hat\Lambda(t) &= \int_0^t \hat\lambda(s) \\
\hat S(t) &= \prod_{s \le t} \left(1 - \frac{\sum w_i dN_i(t)}{\sum_i Y_i(t)}\right) \\
&= \prod_{s \le t} \left(1 - \hat\lambda(s)\right) && (1) \\
FH(t) &= \prod_{s \le t} e^{-lhat(t)} = \exp(-\hat\Lambda(t)) && (2) \\
M_i(t) &= N_i(t) - Y_i(t)\hat\Lambda(t)
\end{aligned}$$

For a multi-state model, let A be the transition matrix.

$$A(t) = \begin{pmatrix} \square & \lambda_{12}(t) & \lambda_{13}(t) & \lambda_{14}(t) & \ldots \\ \lambda_{21}(t) & \square & \lambda_{23}(t) & \lambda_{24}(t) & \ldots \\ \lambda_{31}(t) & \lambda_{32}(t) & \square & \lambda_{34}(t) & \ldots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

The diagonal elements of A as such that row sums are zero. Then we have

$$\begin{aligned}
p(t) &= p_0(t) \prod_{s \le t} \exp(I + A(s)) && (3) \\
&= p_0(t) \prod_{s \le t} \exp(A(s)) && (4)
\end{aligned}$$

Note that (3) is a first order Taylor series for (4).

When $\hat\lambda(t)$ are the simple rates

$$\hat\lambda_{jk}(t) = \frac{\sum_i dN_{ijk}(t)}{\sum_i Y_{ij}(t)}$$

then

- Equation (3) is the Aalen-Johansen estimator.
- If there are only 2 states, it is equal to the Kaplan-Meier
- Essentially no one uses equation (4)

When $\hat{\lambda}_{jk}(t;z)$ is the predicted hazard from a Cox model

$$\hat{\lambda}_{jk}(t;z) = \frac{\sum_i dN_{ijk}(t)}{\sum_i Y_{ij}(t) e^{(X_i - z)\hat{\beta}}}$$

then

- If there are only 2 states
  - Equation (4) is the Breslow estimate of survival
  - No one uses equation (3)
- For ¿ 2 states
  - The "Scandanvian school" all use (3) for theory
  - The mstate library uses (3), but with a caveat
  - The stochastic process literature uses (4)
  - The survival, mshmm, msm packages use (4)

If the prediction is for a subject with large risk and the number at risk at time t is small, then the $I + A$ formula can lead to a negative element in $p(t)$. When this happens, the mstate program fudges the estimate. For great majority of cases, the numeric difference between the two methods will be negligible.

2 Intermittenly observed MSH models

Assume a multi-state hazard model with transition rates between states j and k of $\lambda_{jk}(t)\exp(X\beta)$ For an event based MSH (survival package) this is broken into a baseline hazard and covariate based hazard ratios:

$$r_i(\beta, t) = exp(X_i(t)\beta)$$
$$PL(\beta) = \prod_{jk} \left[\prod_t \frac{\sum_i dN_{ijk}(t) r_i}{\sum_i Y_{ij}(t) r_i} \right]$$

Notice that the hazard does not appear. We can maximize $\hat{\beta}$ and then form estimates of absolute risk $p(t;z)$ for any desired covariate. Key assumptions that are needed to compute the above are

- Over a subject's time at risk, all transitions are observed.

- The time of each transition, and the risk set for that transtion, are known.

In an mshmm or msm model the likelihood is based not on observed *transtions* but rather on observed *states*. A classic framework for this is panel data, where subjects come in a regular intervals and a panel of markers is measured. When a new condition occurs, the inception time of the condition would only be known in an interval.

In the mshmm code, for each trial solution of $\hat\beta$ and each subject i

1. Compute the predicted curve $p(t; x_i)$ for the subject. (For the moment assume the covariate $x_i(t)$ is not time-dependent).
2. Compare the prediction to the observed state for subject i, at each observation time t_{ij} for that subject.
3. Use this to compute an updated $\hat\beta$.

I The notation within the codee replaces $p(t)$ with $\alpha(t)$, a working probabilty. For simple panel data it looks like this. There is a separate computation for each subject.

1. Set $\alpha(0)$ to the starting state for the subject, i.e., (0, 0, ..., 1, 0, 0, ...) where there is a 1 for their observed starting state.
2. $\alpha(t_1) = \alpha(t_0)e^{t_1 A(x, \hat\beta}$ where A is the matrix of rates for this subject. This is provisional.
3. If the subject was observed at time t_1, restrict their $\alpha(t_1)$ by setting to 0 all of the "states where they ain't", i.e., all elements of α except the observed state
4. $\alpha(t_2) = \alpha(t_0)e^{(t_2 - t_1) A(x, \hat\beta}$
5. Restrict $\alpha(t_2)$
6. ...

The final log-likekihood for the subjects is $\log(\sum(\alpha_i))$. Add this up over subjects. Find the $\hat\beta$ that maximizes. We next extend this simple algorithm in 4 ways: flexible baseline hazards and time-dependent covariates, exact observation (death), partially observed states, and hidden Markov models.

The equation $y = exp(At)$ is the solution to the differential equation $dy/dt = A$, i.e., the case of constant transition rates over a time period $(0, t)$. A time dependent covariate needs to be expressed as a series of intervals (t_0, t_1), (t_1, t_2), ... with the covariate as a constant over that interval. A very common time dependent covariate in our models is either current age or current time.

The mshmm and msm functions use an explicit baseline hazard. In the biomedical problems we work with the assumption of a constant baseline hazard, i.e., a single intercept term β_0 is unrealistic. A plot of US life tables shows that for ages over 50, death rates are close to $\exp(\beta_0 + \beta_1 a$ where a is current age. Since the mshmm, msm, and coxph functions all use $\exp(X\beta)$ (it avoids negative hazards, i.e., the dead coming back to life), a linear age term is often a good start, though it will often need to be updated to a spline. Likewise for time since enrollment.

Three centuries of work with national life tables has shown that, except for the very young, yearly death rates are a sufficiently fine division of age, and we will often use integer age as a covariate, it increase by 1 at each birthday. If follow-up for any given subject is fairly short, age at enrollment may sometimes suffice.

A general equation for a given subject looks like below, ignoring the i subscripts; α a vector with one element per state.

$$\alpha = \alpha(t_0)D(t_0,\theta)\, e^{(t_1-t_0)A(t_0,\beta)} D(t_1,\theta)\, e^{(t_2-t_1)A(t_1,\beta)} D(t_1,\theta)\ldots$$

There are types of possible observations at time t in the above

- The hazard matrix at time t will have jk element of $\exp(X_i(t)\beta$ for any transition that can occur, 0 otherwise.
- The forward transtion matrix from a time t_m to t_{m+1} depends on the rate matrix at time t_m and the size of the forward step
- There are 5 types of possible observation at the time point
  - For time points without an observation, e.g., extra times inserted in the data set to advance the age variable, $D = I$, the identity matrix.
  - Interval censored state k was observed. $D_{kk} = 1$ and the remaining elements are 0.
  - A partial state was observed, e.g., the subject is in some set of states a, b, c, ... $D = 0$ except for D_{aa}, D_{bb}, etc. This applies by the way to an observation of “alive”.
  - A death or other exact time was observed, and is state k. $D_{jk} = \lambda_{jk}(t)$ for j/nek, otherwise $D = 0$.
  - A marker variable was observed with value y, then $D_{jj} = f(y|state = j)$

The choices for $\alpha(t_0)$ and $D(t_0)$ can be more involved.